Mohamed Al Mamoun Tantaoui El Araki

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Education

Al Akhawayn University **Fall 2021 – Spring 2026**
Bachelor of Science in Computer Science (BSCSC), Minor in Communication Studies (BACS) *Ifrane, Morocco*

Boğaziçi University **Fall 2024**
Exchange Student - Study Abroad Program in Computer Engineering *Turkey*

Relevant Projects

Government RAG System | *In Development* **Technologies: React, FastAPI, PostgreSQL, MinIO, Ollama, Docker**

- Developing government-grade RAG (Retrieval Augmented Generation) system with complete data sovereignty and zero external API dependencies for secure document analysis and Q&A capabilities.
- Architected microservices system with 100% on-premises processing using Ollama LLM, PostgreSQL vector database, and MinIO storage, ensuring military-grade security with audit logging and access controls.
- Implemented multilingual support (Arabic, French, English) with advanced security guardrails including prompt injection protection, context restriction, and hallucination prevention.
- Currently developing PDF-to-Excel converter feature with drag-and-drop interface, job status tracking, and automated document processing pipeline.

eSaha Mental Health Platform **Technologies: React, TypeScript, Vite, Tailwind CSS, Supabase**

- Developed a comprehensive mental health and wellness platform offering secure user authentication, real-time chat with specialists, and private journaling capabilities.
- Implemented full-stack application using React with TypeScript for frontend, Supabase for backend services (authentication, database, storage), and Tailwind CSS for responsive UI design.
- Built key features including specialists directory, appointment scheduling system, emergency contacts management, and curated mental health resources for scalable user experience.

DCE Web Scraper **Technologies: Python, Selenium WebDriver, Microservices Architecture**

- Architected advanced web scraping system for automated document extraction from public procurement platforms, achieving 95% performance improvement through microservices design.
- Developed parallel processing capabilities with up to 10 concurrent workers, implementing intelligent retry mechanisms and comprehensive error handling for reliable data extraction.
- Refactored monolithic 3,643-line codebase into 7 specialized microservices, improving maintainability and enabling independent scaling of system components.

Quiz Game App | *Concurrent Programming* **Technologies: Kotlin, Firebase, Kotlin Coroutines**

- Developed a mobile app that integrates Firebase for seamless login and signup processes.
- Implemented multi-threading functionality using Kotlin Coroutines to handle concurrent tasks efficiently.
- Designed a real-time question-handling system with a 15-second timer per question, ensuring an interactive and user-friendly interface.

Preparia Learning Platform **Technologies: WordPress, Elementor, H5P**

- Contributed to the development of a web platform to help high school students prepare for university entrance tests in medicine, engineering, and economics.
- Collaborated with the project team to ensure seamless integration of interactive features such as quizzes and videos to improve user engagement.

My Courses App | *In Progress* **Technologies: Flutter, Dart, Android Studio, WordPress, LearnDash LMS**

- Developed a cross-platform mobile application to enable users to browse, register, and access various courses and lessons seamlessly.
- Integrated WordPress as a backend for user registration and course management, leveraging the LearnDash LMS plugin for efficient handling of courses and lessons.

- Designed and implemented a registration page linked to the WordPress site, along with basic login and home page functionalities.
- Planned and initiated a roadmap to include detailed course views, user profile features, search and filter functionality, and UI/UX enhancements in future updates.

Experience

Software Development Intern **June 2025 – August 2025**

Intelcia Solutions

- Developed DCE Scraper, an advanced web scraping system for automating document extraction from public procurement platforms, achieving 95% reduction in processing time (15 minutes vs 2.5 hours for 300 consultations).
- Architected and implemented a microservices system with 7 specialized services, refactoring 3,643-line monolithic codebase into modular components for improved maintainability and scalability.
- Built parallel processing system using Selenium WebDriver with multi-robot support (up to 10 workers), implementing intelligent retry mechanisms and comprehensive error handling to eliminate silent failures.
- Technologies used: Python 3.12+, Selenium WebDriver, YAML configuration management, parallel processing, and async download verification systems.

Project Intern, IN Team **June 2024 – August 2024**

ITS Elaraki School

- Developed a website to support high school students in preparing for competitive exams in medicine, engineering, and economics, using WordPress, Elementor, and H5P for seamless content management and interactive features.
- Enhanced visual elements of the website by improving images and graphics using Adobe Illustrator, contributing to a more engaging user experience.
- Worked on transitioning the website into a mobile application, laying the foundation for future app development and improved user accessibility.

Mentor, AUI Compete Organisation **January 2023 – May 2023**

Al Akhawayn University

- Provided mentorship in academic and personal development to ensure candidates were well-equipped to succeed in the competition.

Technical Skills

Programming Languages & Frameworks: C, Java, JavaScript, TypeScript, Kotlin, Git, SQL, Python, ReactJS, Docker, NextJS, Amazon Web Services, NodeJS
Operating Systems: Windows XP/7/10/11, macOS, Linux, Ubuntu
Languages: English, French, Arabic, Spanish (Beginner)

Tools

Development Tools: VS Code, Android Studio, Git
Design & Media: Adobe Creative Suite (Photoshop, XD, Illustrator, InDesign), Figma, CapCut
Project Management & Productivity: Miro, Jira, Notion, Microsoft Office Suite
Other Tools: UTM, LaTeX